Design and Build a Basic VPC Network Using Terraform

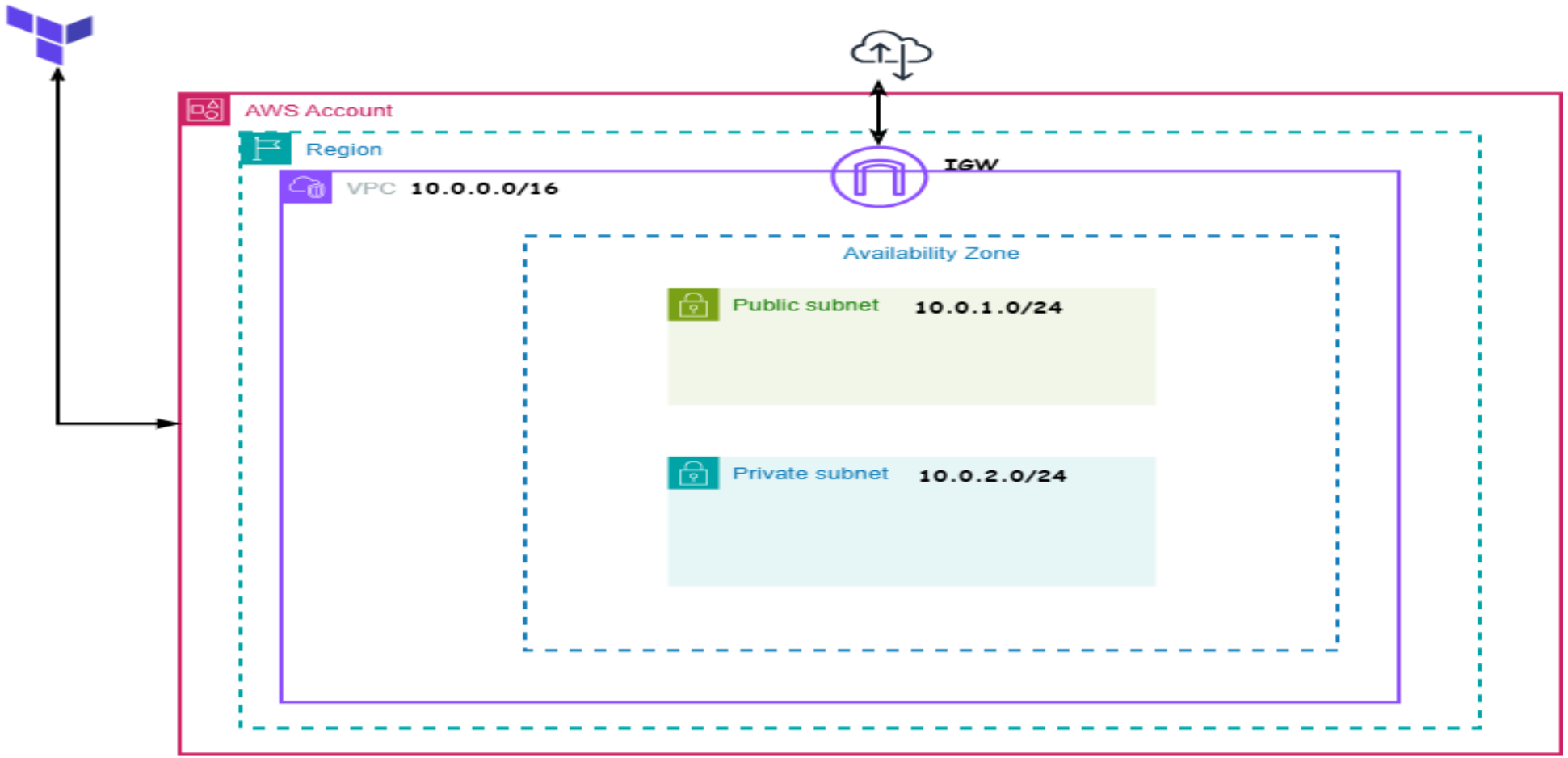
LAB 1B

Task Description

In this lab exercise you will write Terraform configuration files to:

1. Create a **Virtual Private Cloud (VPC)** in AWS
2. Create **one public subnet** within the VPC
3. Create **one private subnet** within the VPC

VPC Network Architecture



Step 1: Create Project Structure

- Open a terminal and create a clean lab directory with the basic files:

```
lab1B/
    ├main.tf
    ├variables.tf
    └ outputs.tf
```

Step 2: Configure Provider

```
terraform {
        required_providers {
                aws = {
                source  = "hashicorp/aws"
                version = "~> 5.0"
                }
        }
}

provider "aws" {
        region = var.region
}
```

Step 3: Define variables

```
variable "region" {
    default = "us-east-1"
}

variable "instance_type" {
    default = "t2.micro"
}
```

Step 4: Write Config for VPC

```
resource "aws_vpc" "main" {
	cidr_block = "10.0.0.0/16"

	tags = {
	Name = "Terraform-VPC"
	}
}
```

Step 5: Write Config for IGW

```
resource "aws_internet_gateway" "igw" {
	vpc_id = aws_vpc.main.id

	tags = {
	Name = "VPC-IGW"
	}
}
```

Step 6: Write Config for public and private subnets

```
resource "aws_subnet" "public" {
        vpc_id = aws_vpc.main.id
        cidr_block = "10.0.1.0/24"
        map_public_ip_on_launch = true

        tags = {
        Name = "Public-Subnet"
        }
}


resource "aws_subnet" "private" {
        vpc_id = aws_vpc.main.id
        cidr_block = "10.0.2.0/24"

        tags = {
        Name = "Private-Subnet"
        }
}
```

Step 7: Write Config for route table

```
resource "aws_route_table" "public_rt" {
        vpc_id = aws_vpc.main.id

        route {
        cidr_block = "0.0.0.0/0"
        gateway_id = aws_internet_gateway.igw.id
        }

        tags = {
        Name = "Public-Route-Table"
        }
}


resource "aws_route_table_association" "public_assoc" {
        subnet_id = aws_subnet.public.id
        route_table_id = aws_route_table.public_rt.id
}
```

Step 8: Write Config for Outputs

```
output "vpc_id" {
value = aws_vpc.main.id
}

output "public_subnet_id" {
  value = aws_subnet.public.id
}

output "private_subnet_id" {
  value = aws_subnet.private.id
}
```

Step 9: Run the project

1. terraform init
2. Terrform fmt
3. Terraform validate
4. terraform plan
5. terraform apply

- 🧹 **Cleanup**

1. terraform destroy

Variable Files and Precedence

- **Passing Variables via Command Line**

This overrides the default value of region.

Syntax

terraform apply -var="key=value"

Example

Passing a Single Variable

terraform apply -var="region=us-west-2"

Passing Multiple Variables

*terraform apply *

*-var="region=us-east-1" *

-var="instance_type=t3.micro"

Variable Files and Precedence

Using a Variable File (.tfvars)

- This overrides the default value of region.
- Create terraform.tfvars file and add these variables

```
region = "us-east-1"
instance_type = "t2.micro“
```

Run:

```
terraform apply -var-file="terraform.tfvars"
```